Fixed-Wing Drone Wind Tunnel Force Measurement Notes – Expanded Build Version

Load Cells, Lift/Drag Force Balance, Mounting, Calibration, and Equations

Isaiah Ramos

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Project context. These notes are for measuring aerodynamic forces on the **FMS 3000 mm Fox** aerobatic EP glider in a wind tunnel. The aircraft has a wingspan of 3.0 m (118.1 in), a wing area of about $0.744\,\mathrm{m}^2$ ($74.4\,\mathrm{dm}^2$, $1154\,\mathrm{in}^2$), a mean chord of about 0.248 m (9.76 in), an aspect ratio of about 12.1, a semi-symmetrical airfoil, and a flying weight of about 4.7 kg (46 N). The goal is to measure **lift** and **drag** using a load-cell force balance.

Expanded revision note. This version adds practical build details that are easy to point to during a presentation: the exact force path, where the hidden load cells physically sit, what bolts to what, a parts list, buying sources, a sizing worksheet, and step-by-step calibration tables. The most important design idea is that the load cells stay outside the airflow while the sting/support beam transmits lift, drag, and moment into the hidden balance.

Contents

1 What Are We Trying to Measure?

When a fixed-wing drone is placed in a wind tunnel, the air exerts forces and moments on it. For a basic test, the two most important forces are:

- **Lift, L:** force perpendicular to the incoming airflow (the relative wind), usually upward.
- **Drag, D:** force parallel to the incoming airflow, pushing the drone backward.

By definition, lift and drag are referenced to the *relative wind*, not to the model's body axes. This distinction matters as soon as the model is pitched: a balance mounted to the tunnel structure measures force along fixed tunnel axes (normal force N, perpendicular to the model reference line, and axial force A, along it), which are not the same as lift and drag unless the angle of attack is zero. The two are related by a rotation through α (see Section 10.3).

A more complete wind-tunnel balance can measure three forces and three moments: normal force, side force, axial force, pitching moment, yawing moment, and rolling moment. NASA describes a wind-tunnel balance as a multi-axis force transducer used to measure aerodynamic loads on a model [1]. For this project, the first practical target is a **two-axis external force balance** that measures lift and drag.

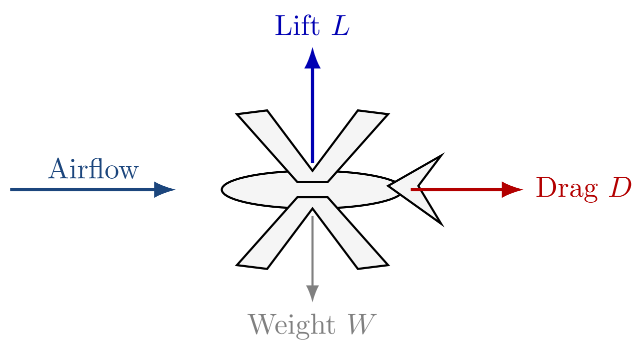


Figure 1: Basic aerodynamic forces on a fixed-wing drone in a wind tunnel.

2 What Is a Load Cell?

A **load cell** is a force sensor. Most small electronic load cells are **strain-gauge load cells**. They convert a mechanical force into a small electrical signal.

2.1 Simple Explanation

A load cell is usually a metal beam or block with strain gauges attached to it. When force is applied, the metal bends by a very small amount. This bending stretches or compresses the strain gauges. The strain gauges change electrical resistance slightly, and the electronics measure that change.

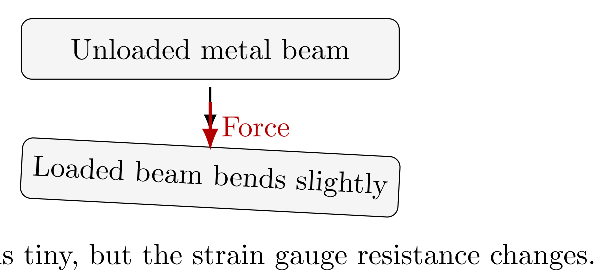


Figure 2: A load cell measures force by sensing tiny bending/strain in a metal structure.

2.2 Important Concept: A Load Cell Measures Along Its Mounted Axis

A load cell does not automatically know whether it is measuring lift or drag. It measures force along the direction it is mounted and loaded. Therefore:

- A **vertical load cell** can measure lift.
- A **horizontal load cell** can measure drag.
- A bad mount can mix lift into the drag reading and drag into the lift reading.

3 How a Strain-Gauge Load Cell Works

3.1 Strain

Strain is the fractional change in length of a material:

$$\varepsilon = \frac{\Delta L}{L} \tag{1}$$

where:

- ε = strain, unitless
- ΔL = change in length
- L = original length

In a load cell, the strain is extremely small. The metal beam may bend only a tiny amount, but strain gauges are sensitive enough to detect it.

3.2 Gauge Factor

A strain gauge changes resistance when stretched or compressed. The **gauge factor** relates fractional resistance change to strain:

$$GF = \frac{\Delta R/R}{\varepsilon} \tag{2}$$

Rearranged:

$$\frac{\Delta R}{R} = GF \cdot \varepsilon \tag{3}$$

where:

- GF = gauge factor
- R = nominal strain gauge resistance
- ΔR = change in resistance

3.3 Wheatstone Bridge

The resistance change is very small, so load cells usually wire the strain gauges into a **Wheatstone bridge**. SparkFun notes that many load cells use a standard four-wire Wheatstone bridge configuration when connected to HX711 amplifier boards [6].

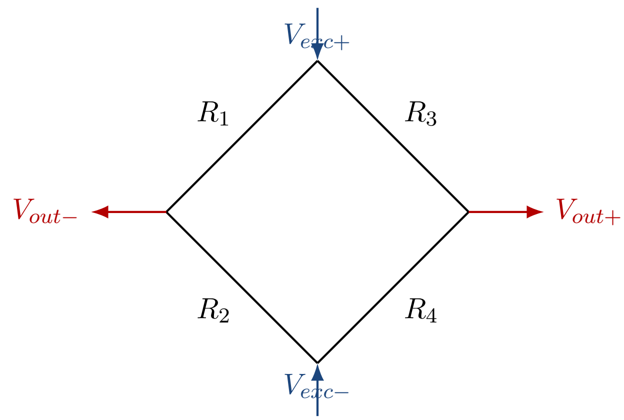


Figure 3: Simplified Wheatstone bridge. Small resistance changes create a small differential output voltage.

A general Wheatstone bridge output can be written as:

$$V_{out} = V_{exc} \left(\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right) \quad (4)$$

where:

- V_{out} = small bridge output voltage
- V_{exc} = excitation voltage supplied to the bridge
- R_1, R_2, R_3, R_4 = bridge resistances

When no force is applied, the bridge is approximately balanced and V_{out} is near zero. When force bends the load cell, the resistances change and V_{out} becomes positive or negative depending on the force direction.

Full-bridge detail. The equation above is the general expression for any Wheatstone bridge. A real strain-gauge load cell is usually a *full bridge*: all four arms are active gauges, bonded so that two stretch (resistance up) and two compress (resistance down) under load. Wiring them this way roughly quadruples the output compared with a single active gauge and cancels most temperature drift, because all four gauges sit on the same metal and shift together with temperature. This is why a load cell gives a clean, repeatable signal that a single loose strain gauge cannot.

4 Why an Amplifier/ADC Is Needed

The load-cell signal is very small, often in the millivolt range. A normal Arduino analog input is not good enough for accurate force measurement. A load-cell amplifier/ADC board, such as the HX711, is commonly used to read the tiny signal from a load cell. SparkFun describes the HX711 as a board used to get measurable data from a load cell and strain gauge [5].

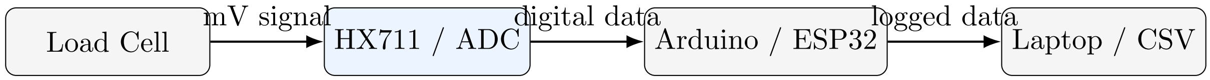


Figure 4: Typical signal chain for load-cell measurement.

5 Wind-Tunnel Force Balance Setup

The correct wind-tunnel setup is not to glue a pressure pad to the wing. Instead, the drone is mounted to a rigid support called a **sting**. The sting transfers the aerodynamic loads into a force balance. NASA shows external wind-tunnel balances where the model is mounted in the test section and the aerodynamic forces are determined from balance/gauge readings [2].

5.1 External Two-Axis Balance

A simple two-axis balance uses one load cell for lift and one load cell for drag.

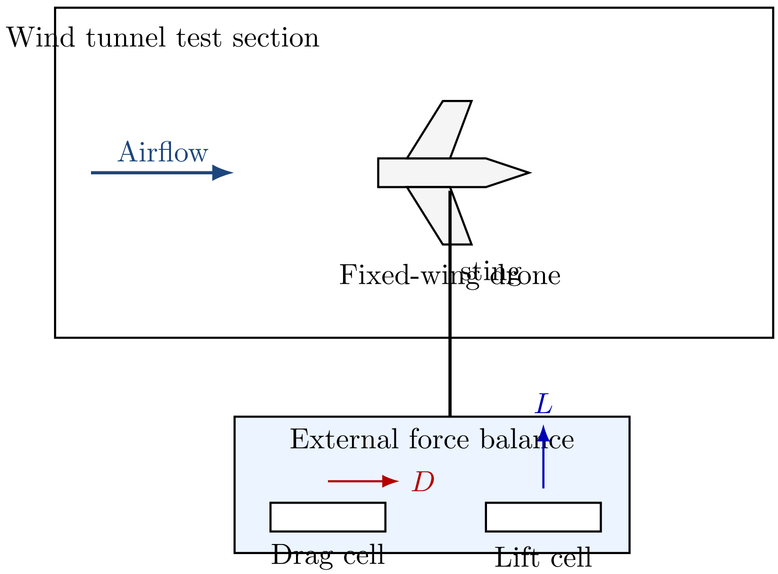


Figure 5: Conceptual external force-balance setup for measuring lift and drag.

5.2 What Each Sensor Measures

Table 1: Sensor axis definitions for a two-axis balance.

Axis	**Load Cell Direction**	**Main Measurement**
Lift axis	Vertical	Upward/downward aerodynamic force
Drag axis	Horizontal, parallel to airflow	Backward aerodynamic force
Pitching moment	Rotation about lateral axis	Nose-up/nose-down torque

Important: A two-axis balance can measure lift and drag, but a fixed-wing drone also produces pitching moment. If the drone is large or tested across many angles of attack, adding pitching moment measurement is strongly recommended.

6 Where Are the Load Cells Actually Located?

In professional wind-tunnel photos, the load cells are usually not visible. The visible part is the **sting**, post, or support beam. The actual sensing hardware is hidden inside the model, inside the sting, behind a fairing, inside the floor mount, or below the wind-tunnel floor. This is intentional: the sensors should measure the forces transmitted from the aircraft, not the aerodynamic drag of the sensors themselves.

Core idea. The load cells are hidden from the airflow. The support sting carries the lift, drag, and moment loads from the drone into the hidden force balance.

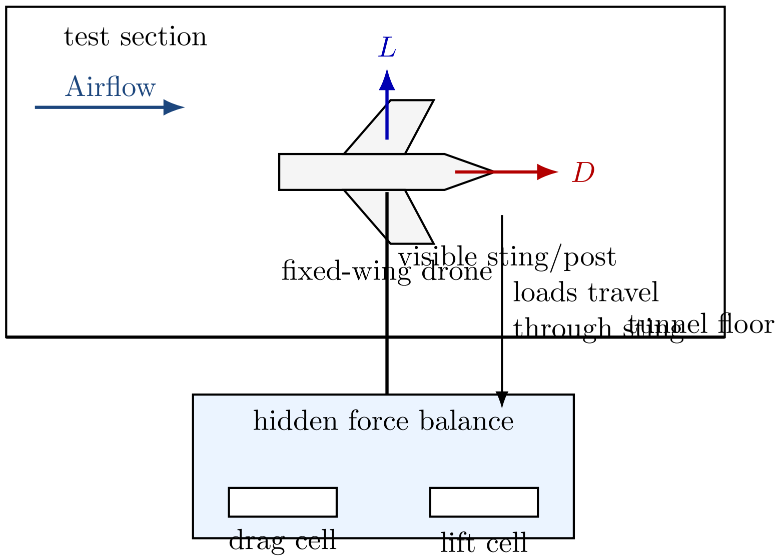


Figure 6: The load cells should be hidden outside the main airflow. The sting transfers the forces into the hidden force balance.

6.1 Force Path Through the Mount

The actual load path is:

1. Airflow pushes on the drone.
2. The drone pushes/pulls on the mounting point.
3. The sting transfers that force to the balance frame.
4. The balance frame loads the vertical load cell, horizontal load cell, and any moment-measurement cells.
5. The electronics convert the load-cell signal into force readings.

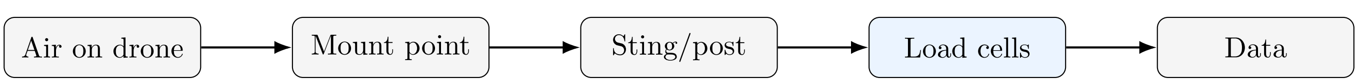


Figure 7: Force path from aerodynamic load to recorded data.

6.2 Why the Load Cells Should Not Be Exposed

If the load cells, wires, brackets, or amplifier boards are exposed to the tunnel flow, they create their own drag. The balance cannot tell the drone's drag apart from the hardware's drag, so the measured axial force becomes the sum of every drag-producing object in the flow plus their mutual interference:

$$D_{measured} = D_{drone} + D_{sensor} + D_{wires} + D_{brackets} + D_{interference} \tag{5}$$

This is bad because the sensor system contaminates the aerodynamic data, and the contamination is not a fixed offset you can simply subtract: it changes with speed and angle of attack. The cleaner arrangement keeps everything except the drone and a thin support out of the flow:

$$D_{measured} \approx D_{drone} + D_{sting\ interference} \tag{6}$$

The sting still causes some interference, but it is much easier to minimize, hold constant, and document than an exposed sensor block inside the tunnel. A common practice is to also run the sting alone (no model, or with a dummy body) to characterize and subtract its contribution.

7 Key Aerodynamic Equations

7.1 Dynamic Pressure

Dynamic pressure is:

$$q = \frac{1}{2}\rho V^2 \tag{7}$$

where:

- q = dynamic pressure, Pa or N/m^2
- ρ = air density, approximately 1.225 kg/m^3 near sea level
- V = wind-tunnel airspeed, m/s

7.2 Lift Equation

NASA gives the lift equation as lift coefficient times density times half the velocity squared times wing area [3]:

$$L = C_L \left(\frac{1}{2}\rho V^2\right) S \tag{8}$$

or:

$$L = C_L q S \tag{9}$$

7.3 Drag Equation

NASA gives the drag equation as drag coefficient times density times half the velocity squared times reference area [4]:

$$D = C_D \left(\frac{1}{2}\rho V^2\right) S \tag{10}$$

or:

$$D = C_D q S \tag{11}$$

7.4 Lift and Drag Coefficients

After measuring force, solve for the coefficients:

$$C_L = \frac{L}{qS} \tag{12}$$

$$C_D = \frac{D}{qS} \tag{13}$$

These coefficients are often more useful than raw force because they make it easier to compare results at different wind speeds or model sizes. One caveat: C_L and C_D are only directly comparable between cases at the same *Reynolds number*, $Re = \rho V c/\mu$, where c is a reference length (usually the chord) and μ is air viscosity. Re sets how the boundary layer behaves, so the same wing tested slowly versus quickly, or a small model versus a full-size one, can show measurably different C_D (and a different stall angle) even at identical α. Report the test Re alongside any coefficient. For the Fox, using the mean chord $c \approx 0.248\,\mathrm{m}$ and standard air ($\rho = 1.225\,\mathrm{kg/m^3}$, $\mu \approx 1.81 \times 10^{-5}\,\mathrm{Pa\,s}$), the chord Reynolds number is about 170 000 at 10 m/s, 250 000 at 15 m/s, and 340 000 at 20 m/s. This is a low-Reynolds regime where airfoil behavior can be sensitive to surface finish and laminar-separation effects, so it is worth recording carefully.

8 Drone Size and Force Estimate

The aircraft is the FMS 3000 mm Fox. Its wingspan is a measured value, not an estimate:

$$b = 3000\,\mathrm{mm} = 3.0\,\mathrm{m} = 118.1\,\mathrm{in} \tag{14}$$

Common mix-up. The 56 in figure sometimes quoted for this model is the length of the carbon wing *spar tube*, not the half-span. The true half-span is about 59 in and the full span is 118.1 in (3.0 m). Use the full span for area and force calculations.

The wing reference area and mean chord are also known from the manufacturer data:

$$S \approx 0.744\,\mathrm{m}^2 \quad (74.4\,\mathrm{dm}^2,\ 1154\,\mathrm{in}^2) \tag{15}$$

$$c_{avg} = \frac{S}{b} = \frac{0.744}{3.0} \approx 0.248\,\mathrm{m} \approx 9.76\,\mathrm{in} \tag{16}$$

The aspect ratio is high, as expected for a glider:

$$AR = \frac{b^2}{S} = \frac{3.0^2}{0.744} \approx 12.1 \tag{17}$$

Flying weight and stall speed. The model weighs about 4.7 kg, so its weight is $W = mg \approx 46\,\mathrm{N}$. In steady level flight the wing must produce lift equal to weight, $L = W$. Using a semi-symmetrical-airfoil estimate $C_{L,max} \approx 1.2$, the stall speed is

$$V_{stall} = \sqrt{\frac{2W}{\rho S C_{L,max}}} = \sqrt{\frac{2(46)}{(1.225)(0.744)(1.2)}} \approx 9.2\,\mathrm{m/s}. \tag{18}$$

This matters for interpreting the force tables below: at low speed the lift is only tens of Newtons (it must match the 46 N weight near stall), and the large lift figures appear only at high tunnel speed and high angle of attack, where the wing is loaded far beyond 1 g.

8.1 Example Force Estimate

Assume:

- Full span: $b = 3.0\,\mathrm{m}$
- Average chord: $c_{avg} = 0.248\,\mathrm{m}$
- Wing area: $S \approx 0.744\,\mathrm{m}^2$
- Wind speed: $V = 15\,\mathrm{m/s}$
- Air density: $\rho = 1.225\,\mathrm{kg/m}^3$
- Example lift coefficient: $C_L = 0.8$
- Example drag coefficient: $C_D = 0.08$

Dynamic pressure:

$$q = \frac{1}{2}(1.225)(15^2) \approx 138\,\mathrm{Pa} \tag{19}$$

Lift estimate:

$$L = qSC_L = (138)(0.744)(0.8) \approx 82\,\mathrm{N} \tag{20}$$

Drag estimate:

$$D = qSC_D = (138)(0.744)(0.08) \approx 8.2\,\mathrm{N} \tag{21}$$

Note on coefficient assumptions. This example uses a moderate cruise condition ($C_L = 0.8$, $C_D = 0.08$) to illustrate typical operating forces. For *sizing the load cells*, a more conservative high-angle-of-attack condition is used later in the notes ($C_{L,max} \approx 1.2$, $C_{D,max} \approx 0.18$), which produces larger forces. The two cases are not contradictory: cruise estimates show typical readings, while max-load estimates set the sensor capacity. A clean glider like the Fox actually cruises at a lower C_D (roughly 0.03 to 0.05), so 0.08 here is already conservative; always size hardware against the larger case.

This example shows why the lift load cell must usually have a higher capacity than the drag load cell.

9 Load-Cell Capacity: What Does “5 kg” Mean?

Load cells are often labeled in kilograms, such as 5 kg, 10 kg, or 20 kg. This usually means kilogram-force capacity, not mass for an aerodynamic equation. Convert to Newtons using:

$$F = mg \tag{22}$$

where $g \approx 9.81\,\mathrm{m/s^2}$.

$$1\,\mathrm{kgf} \approx 9.807\,\mathrm{N} \tag{23}$$

Table 2: Common load-cell capacities converted to Newtons (using $1\,\mathrm{kgf} = 9.807\,\mathrm{N}$).

Load Cell Label	Approx. Max Force	Best Use
1 kg	9.81 N	Small drag forces, very sensitive but easy to overload
5 kg	49.0 N	Medium drag or small lift measurements
10 kg	98.1 N	Larger drag or moderate lift
20 kg	196 N	Large lift force for bigger models
50 kg	490 N	High lift force or high-speed testing

9.1 Recommended Starting Range for This Drone

Because the drone has a full wingspan of 3.0 m, the lift force can become significant at high tunnel speed. A reasonable first estimate is:

Table 3: Suggested load-cell ranges for the current fixed-wing drone.

Axis	Suggested Range	Reason
Lift	20 kg to 50 kg	Lift can be tens to over 100 N depending on speed and chord
Drag	5 kg to 10 kg	Drag is usually much smaller than lift
Pitching moment	two vertical cells or torque sensor	Useful for nose-up/nose-down stability data

Do not oversize too much. A 50 kg load cell may survive large lift loads, but it may be less sensitive to small forces. Pick a capacity with a safety margin, but not a massive one. A common starting point is roughly **2x to 3x** the expected maximum force.

10 Mounting Methods

10.1 Belly Sting Mount

A belly sting mount supports the drone from underneath.

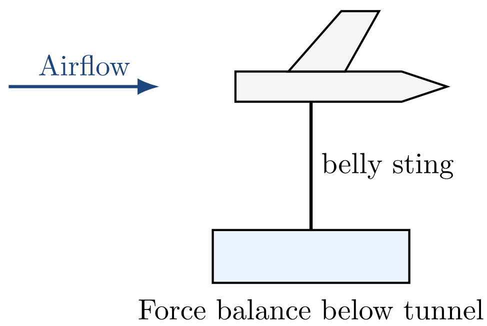


Figure 8: Belly sting mount. Easier to build, but it can disturb airflow under the fuselage and wing.

Pros:

- Easier to build.
- Easier to adjust angle of attack.
- Good for early student testing.

Cons:

- Support can disturb airflow under the drone.
- Mount can create extra drag.
- Must be rigid enough to avoid vibration.

10.2 Rear Sting Mount

A rear sting supports the drone from behind.

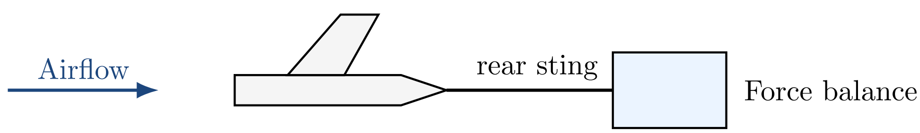


Figure 9: Rear sting mount. Cleaner for some tests, but harder to attach and may affect tail flow.

Pros:

- Can reduce interference with wing underside.
- Common style in wind-tunnel testing.

Cons:

- Harder to attach to a complete drone.
- Can affect tail and propeller-region flow.
- More mechanically difficult to align.

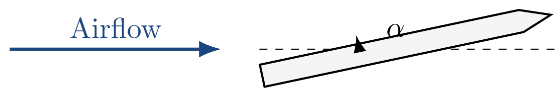
10.3 Angle-of-Attack Adjustment

Angle of attack, usually written α, is the angle between the incoming airflow and a reference line on the drone or wing.

$$\alpha = \text{angle between wing/body reference line and incoming airflow} \tag{24}$$

A good mount should allow testing at several angles, for example:

$$\alpha = -5^\circ,\ 0^\circ,\ 5^\circ,\ 10^\circ,\ 15^\circ,\ 20^\circ \tag{25}$$



Angle of attack changes lift, drag, and pitching moment.

Figure 10: Angle of attack should be adjustable and measured carefully.

Resolving Measured Force into Lift and Drag

If the balance is built into the model and rotates with it (the usual student arrangement, where the lift cell stays "vertical relative to the model" and the drag cell stays "along the model"), then at nonzero α the cells do *not* read lift and drag directly. They read **normal force** N (perpendicular to the model reference line) and **axial force** A (along it). Lift and drag, which are referenced to the relative wind, are recovered by rotating through α:

$$L = N\cos\alpha - A\sin\alpha \tag{26}$$

$$D = N\sin\alpha + A\cos\alpha \tag{27}$$

At $\alpha = 0$ these reduce to $L = N$ and $D = A$, which is why the distinction is easy to forget. But at $\alpha = 15^\circ$, ignoring the rotation puts roughly a 1/4 of the (large) normal force into the wrong axis, which can swamp the (small) drag reading entirely. There are two ways to avoid the error:

- **Rotate the data:** keep the balance fixed to the model, record N and A, and apply the equations above using the known α.
- **Rotate the model only:** mount the balance to the fixed tunnel frame so its axes always stay aligned with the wind, and pitch only the model relative to the balance through the angle-of-attack bracket. Then the cells read true lift and drag at every α with no rotation needed. This is cleaner and is the recommended approach for this build.

11 Calibration

Calibration converts raw sensor values into real forces in Newtons.

11.1 Basic Calibration Equation

A simple linear calibration is:

$$F = aR + b \tag{28}$$

where:

- F = force in Newtons
- R = raw ADC reading
- a = calibration slope
- b = offset

11.2 Lift Calibration

To calibrate lift, apply known vertical loads using known masses:

$$F = mg \tag{29}$$

Examples:

Table 4: Known masses for calibration.

Mass	**Force**
0.1 kg	0.981 N
0.2 kg	1.96 N
0.5 kg	4.91 N
1.0 kg	9.81 N
5.0 kg	49.0 N

11.3 Drag Calibration

For drag calibration, apply a known horizontal force using a string, pulley, and hanging mass.

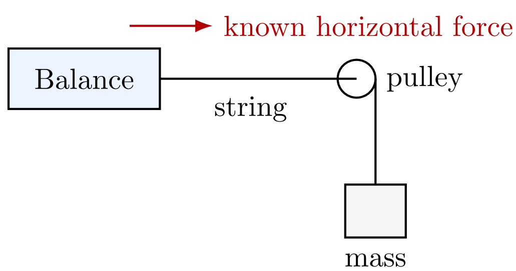


Figure 11: Drag calibration using a string, pulley, and known hanging mass.

If the hanging mass is m, the horizontal force is approximately:

$$F_{drag,cal} = mg \tag{30}$$

This assumes the pulley is low-friction and the string is aligned with the drag axis.

12 Tare, Offsets, and Wind-On Measurement

The load cells will not read zero just because there is no wind. They may measure the weight of the drone, preloads in the mount, cable forces, and alignment errors. Therefore, the wind-off reading must be subtracted.

12.1 Wind-Off Tare

With the tunnel off:

$$L_{zero} = \text{lift-channel reading with no wind} \tag{31}$$

$$D_{zero} = \text{drag-channel reading with no wind} \tag{32}$$

With the tunnel on:

$$L_{on} = \text{lift-channel reading with wind on} \tag{33}$$

$$D_{on} = \text{drag-channel reading with wind on} \tag{34}$$

Actual aerodynamic forces:

$$L = L_{on} - L_{zero} \tag{35}$$

$$D = D_{on} - D_{zero} \tag{36}$$

Practical rule: Tare the sensors at every angle of attack before turning on the tunnel. Changing angle of attack can change how the model weight loads the sensors.

13 Cross-Axis Coupling

Cross-axis coupling happens when a force in one direction appears in the wrong sensor channel. For example, applying pure lift may accidentally change the drag sensor reading.

$$\text{Pure lift applied} \quad \Rightarrow \quad \text{drag channel also changes slightly} \tag{37}$$

A simple correction can be written as a matrix:

$$\begin{bmatrix} L \\ D \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} R_L \\ R_D \end{bmatrix} \tag{38}$$

where:

- R_L = raw lift-channel reading
- R_D = raw drag-channel reading
- L = corrected lift
- D = corrected drag

For a beginner setup, at least check coupling manually:

1. Apply a known vertical load.
2. Record both lift and drag channel changes.
3. Apply a known horizontal load.
4. Record both lift and drag channel changes.
5. Use those results to estimate correction factors.

14 Detailed Parts List and Where to Buy

This section is a practical bill of materials. Exact product choice depends on budget and required accuracy. Avoid relying on only one vendor; load cells and bridge amplifiers are standard parts sold by robotics, electronics, and test-instrumentation suppliers.

14.1 Minimum Working System

A basic lift/drag measurement system needs the following:

Table 5: Minimum parts needed for a two-axis lift/drag balance.

Part	**Purpose**	**Notes**
Lift load cell	Measures vertical aerodynamic force	Usually larger capacity than drag; likely 20 kg to 50 kg for this drone depending on tunnel speed
Drag load cell	Measures horizontal aerodynamic force	Usually smaller capacity than lift; likely 5 kg to 10 kg
Load-cell amplifier/ADC	Converts tiny bridge signal to digital data	HX711 is cheap and fine for steady readings; Phidgets/NI systems are easier or faster for lab data
Arduino/ESP32 or laptop DAQ	Logs force data	Arduino/ESP32 is low cost; USB DAQ is cleaner for lab reports
Sting or support post	Transfers drone loads into force balance	Should be rigid, smooth, and hidden/minimized in airflow
Angle-of-attack mount	Sets drone angle during testing	Use a protractor plate, inclinometer, or indexed holes
Pitot tube or tunnel speed sensor	Measures airspeed V	Needed to compute q, C_L, and C_D
Calibration masses and pulley	Applies known loads	Used to calibrate lift and drag channels
Rigid base frame	Keeps everything aligned	Aluminum extrusion, steel plate, or machined/printed reinforced structure

14.2 Where to Buy: Budget, Midrange, and Lab-Grade Options

Table 6: Suggested suppliers and part categories. Check current stock and datasheets before ordering.

Supplier	Useful Parts	When to Use
SparkFun	HX711 board, NAU7802/Qwiic Scale board, small straight-bar load cells	Best for a low-cost Arduino prototype. SparkFun's HX711 board is specifically intended to interface load cells with microcontrollers [6]. SparkFun also sells small 5 kg straight-bar load cells [7].
Phidgets	Single-point load cells, Wheatstone Bridge Phidget, 4-input PhidgetBridge	Good middle option when you want USB logging and less custom electronics. The PhidgetBridge 4-Input can read up to four load cells and convert readings using calibration parameters [9].
FUTEK	S-beam, miniature tension/compression cells, instrumentation	Better for serious test stands where the force reverses direction or where overload protection matters. FUTEK's S-beam documentation emphasizes in-line tension/compression loading and proper installation [15].
Omega / DwyerOmega	Compression cells, S-beam cells, miniature universal cells, calibrated sensors	Lab/industrial option when accuracy, ruggedness, and calibration documentation matter. Omega lists load cells across compression, S-beam, miniature universal, and beam styles [12].
DigiKey / Mouser	Amplifier boards, ADCs, connectors, cable, strain-gauge electronics	Good for electronics sourcing, especially if the exact breakout board is unavailable. DigiKey lists SparkFun's HX711 amplifier as a sensor/transducer amplifier part [13].
Matek / Holybro / RC autopilot vendors	Pitot tube and MS4525DO airspeed sensor	Useful for measuring tunnel speed or onboard airspeed. The Matek ASPD-4525 uses a TE MS4525DO differential pressure sensor and outputs I2C data [14].
McMaster-Carr / 80-20 / local machine shop	Aluminum extrusion, steel plate, bearings, pulleys, fasteners, threaded rod	Mechanical hardware for the balance frame, sting, angle plate, and calibration pulley system.

Important buying advice. Cheap straight-bar scale load cells can work for a prototype, but for a real wind-tunnel balance, a bidirectional tension/compression load cell or an S-beam style load cell is usually easier to mount correctly. The more serious the test, the more important proper calibration, overload rating, and in-line loading become.

14.3 Recommended Purchase Paths

Lowest-cost prototype path:

- 1x 20 kg to 50 kg load cell for lift.
- 1x 5 kg to 10 kg load cell for drag.

- 2x HX711 or NAU7802 amplifier boards.
- Arduino/ESP32.
- Calibration weights, string, pulley, and rigid frame hardware.

Cleaner student-lab path:

- 2–3x better quality tension/compression load cells.
- PhidgetBridge 4-Input or similar multi-channel bridge DAQ.
- Laptop logging directly to CSV.
- More rigid aluminum/steel balance frame.
- Pitot/differential-pressure sensor for tunnel speed.

Best measurement path:

- 3-component external balance: lift, drag, and pitching moment.
- Calibrated S-beam or miniature tension/compression cells.
- Multi-channel bridge DAQ with simultaneous readings.
- Proper calibration matrix for cross-axis coupling.
- Machined metal frame or carefully designed flexure balance.

15 How to Determine Which Load Cell You Need

The correct load-cell capacity is determined from the largest force you expect during testing, plus tare/preload and safety margin. Do **not** simply choose the largest load cell available. A huge load cell may survive, but it will have worse resolution for small drag forces.

15.1 Step 1: Measure the Drone Geometry

Measure:

- Total wingspan b.
- Average chord c_{avg}.
- Wing reference area $S \approx bc_{avg}$.
- Maximum wind tunnel speed V_{max}.
- Expected angle-of-attack range.

For the current drone:

$$b = 3.0\,\mathrm{m} = 118.1\,\mathrm{in} \tag{39}$$

The actual published chord and area are known for this aircraft ($c_{avg} \approx 0.248\,\mathrm{m}$, $S \approx 0.744\,\mathrm{m}^2$), so the measured values should be used directly. The table below is shown only to illustrate how area scales with chord for a 3.0 m span; the 10 in row brackets the true value.

Table 7: Wing-area estimates for a 3.0 m (118.1 in) span, by assumed chord.

Average Chord	**Chord in meters**	**Approx. Wing Area** S
8 in	0.203 m	0.610 m^2
10 in	0.254 m	0.762 m^2
12 in	0.305 m	0.914 m^2
14 in	0.356 m	1.067 m^2

15.2 Step 2: Estimate Maximum Lift and Drag

Use conservative coefficient estimates. For early sizing, use:

$$C_{L,max,estimate} \approx 1.0 \text{ to } 1.4 \tag{40}$$

$$C_{D,max,estimate} \approx 0.06 \text{ to } 0.10 \quad \text{(efficient high-aspect-ratio glider)} \tag{41}$$

The drag-coefficient range above is for a clean, high-aspect-ratio glider like the Fox, where profile drag is low (≈ 0.025) and the total is dominated by induced drag at high lift. A draggy powered model or one with fixed landing gear and a blunt fuselage could reach $C_D \approx 0.15$ to 0.25; do not use those higher numbers for this aircraft or you will oversize the drag cell. Then calculate:

$$L_{max} = \frac{1}{2}\rho V_{max}^2 S C_{L,max} \tag{42}$$

$$D_{max} = \frac{1}{2}\rho V_{max}^2 S C_{D,max} \tag{43}$$

15.3 Step 3: Add Safety Factor

A practical selection rule is:

$$F_{rated} \geq SF\left(F_{aero,max} + F_{tare} + F_{possible\ misalignment}\right) \tag{44}$$

where SF is a safety factor. For a student wind-tunnel balance, a reasonable starting point is:

$$SF = 2 \text{ to } 3 \tag{45}$$

Load-cell selection guides generally emphasize choosing capacity based on live load, dead/tare load, direction of load, overload risk, and safety margin [17]. Some load-cell datasheets also specify a separate **safe overload**; for example, FUTEK's LSB200 documentation lists overload protection in tension and compression for certain configurations [16]. Never design the normal operating load to depend on the overload rating.

15.4 Step 4: Convert Newtons to kgf

Many load cells are sold in kg or lb capacity. Convert force to kilogram-force:

$$F_{kgf} = \frac{F_N}{9.807} \tag{46}$$

Example:

$$150\,\text{N} \div 9.807 \approx 15.3\,\text{kgf} \tag{47}$$

With a safety factor, this may push the lift load cell selection toward 20 kg, 30 kg, or 50 kg.

15.5 Step 5: Check Resolution

If the load cell is too large, the drag readings may be noisy or low-resolution. Resolution depends on the sensor output, amplifier, noise, and mechanical vibration, but the practical rule is:

> Choose the smallest load-cell capacity that safely survives the largest expected load, including tare, misalignment, and safety factor.

For this drone, a reasonable starting choice is:

Table 8: Initial load-cell choice for the current drone.

Channel	**Starting Capacity**	**Why**
Lift	20 kg to 50 kg	Large wingspan can create tens to hundreds of Newtons at moderate tunnel speed. Choose closer to 20 kg for lower speeds; closer to 50 kg for higher speeds or high angle of attack.
Drag	5 kg to 10 kg	Drag is usually much smaller than lift. A smaller drag cell gives better sensitivity.
Pitch moment	Two lift cells or torque cell	If using two vertical cells, size each for at least half the expected lift plus safety margin.

15.6 Sizing Example at $20\,\mathrm{m/s}$

Assume:

- $b = 3.0\,\mathrm{m}$
- $c_{avg} = 0.248\,\mathrm{m}$
- $S = 0.744\,\mathrm{m}^2$
- $V_{max} = 20\,\mathrm{m/s}$
- $C_{L,max} = 1.2$
- $C_{D,max} = 0.08$ (clean glider profile drag of ≈ 0.025 plus induced drag $C_L^2/(\pi\, AR\, e) \approx 0.047$ at $C_L = 1.2$, $AR \approx 12$)
- $\rho = 1.225\,\mathrm{kg/m^3}$

Dynamic pressure:

$$q = \frac{1}{2}(1.225)(20^2) = 245\,\mathrm{Pa} \tag{48}$$

Lift:

$$L_{max} = (245)(0.744)(1.2) \approx 219\,\mathrm{N} \tag{49}$$

Drag:

$$D_{max} = (245)(0.744)(0.08) \approx 14.6\,\text{N} \tag{50}$$

Convert to kgf:

$$L_{max} \approx \frac{219}{9.807} = 22.3\,\text{kgf} \tag{51}$$

$$D_{max} \approx \frac{14.6}{9.807} = 1.5\,\text{kgf} \tag{52}$$

With safety margin, this example suggests approximately:

- Lift channel: 50 kg if testing up to 20 m/s and high angle of attack. Note that the aircraft's own weight (≈ 4.7 kg, 46 N) is carried by the lift cells as a tare load before any wind, so the cells must comfortably hold aerodynamic lift *plus* model weight.
- Drag channel: 5 kg is enough at 20 m/s (drag is only $\approx$ 15 N, 1.5 kgf); a 10 kg cell adds margin for higher-speed runs while still giving usable sensitivity. Because this is an efficient glider, its drag is small, so do not oversize the drag channel.

If the tunnel speed is lower, such as 10 m/s, the forces drop by a factor of four because force scales with V^2. Note also that the Fox stalls near 9 m/s in free flight at this wing loading, so test speeds below about 10 m/s sit close to or below stall and are mainly useful for low-load calibration checks rather than clean attached-flow data.

15.7 Simple Decision Flowchart

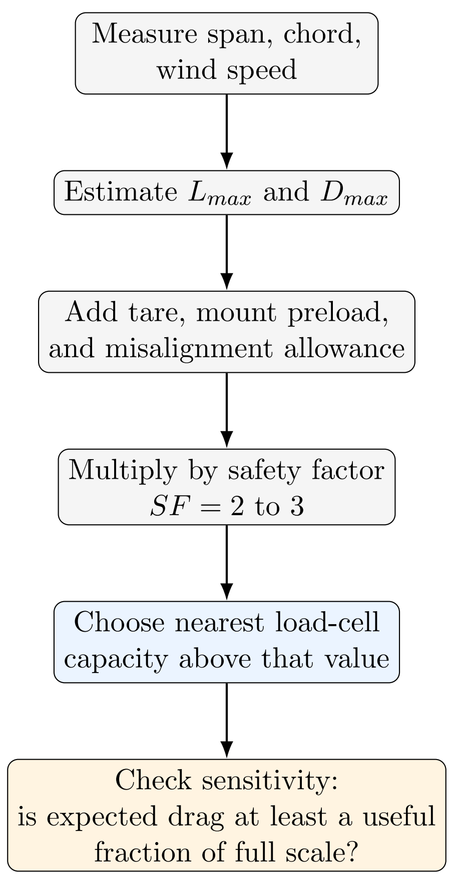


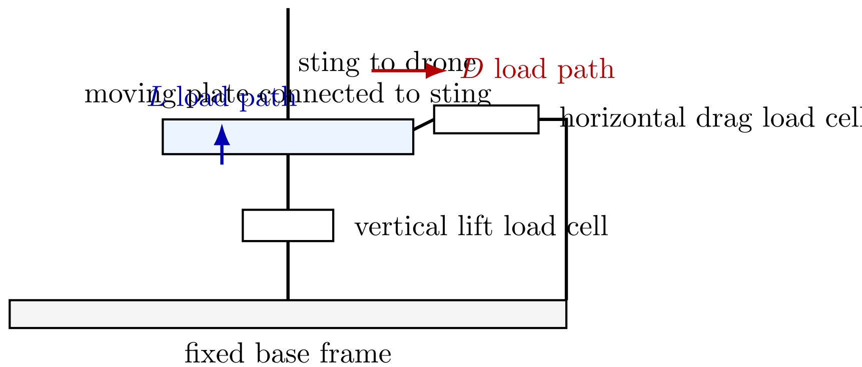
Figure 12: Load-cell sizing process.

16 Practical Mechanical Setup: How to Build the Balance

The cleanest student setup is a hidden external balance under the wind-tunnel floor or behind the test section. The drone is connected to the balance through a smooth sting. The load cells should be mounted so the force enters them along their intended load axis.

16.1 Two-Axis Balance Layout

A practical two-axis layout uses a moving plate connected to the drone sting. The moving plate is restrained by a vertical load cell and a horizontal load cell.



The moving plate barely deflects. The load cells sense the force while the base frame provides the fixed reference.

Figure 13: Simplified two-axis external balance. The load cells are outside the flow and the sting sends the loads into this structure.

16.2 Three-Cell Layout for Lift, Drag, and Pitching Moment

For a large fixed-wing drone, a better setup uses two vertical load cells spaced apart plus one horizontal drag load cell. The two vertical cells provide total lift and pitching moment.

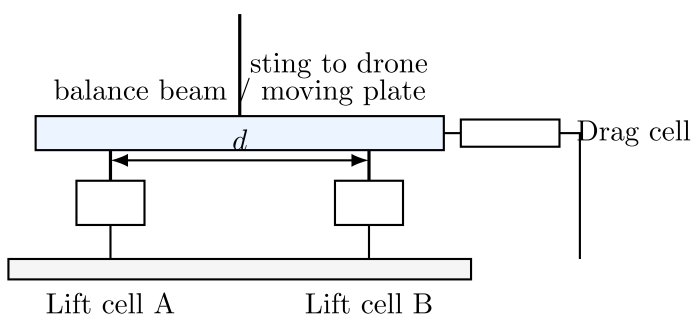


Figure 14: Three-cell setup. Lift is the sum of two vertical cells, while pitching moment comes from their difference.

For this layout, with F_A the front (upstream) vertical cell and F_B the rear (downstream) vertical cell:

$$L = F_A + F_B \tag{53}$$

If the two vertical cells are separated by distance d, an approximate pitching moment about the midpoint is:

$$M_{pitch} \approx (F_B - F_A)\,\frac{d}{2} \tag{54}$$

Sign convention. The same convention $M_{pitch} \approx (F_B - F_A)\, d/2$ is used everywhere in these notes, where F_A is the upstream cell and F_B the downstream cell. A positive result indicates more load on the downstream cell (a nose-down tendency about the midpoint). The exact moment equation depends on where the desired moment reference point is located, such as the quarter chord or the drone center of gravity; shift d accordingly if you reference the moment to a point other than the midpoint.

16.3 Mechanical Build Steps

1. **Build a fixed base.** Use a rigid plate, aluminum extrusion, or steel frame. The base must not flex much under load.
2. **Add the moving plate.** This is the plate connected to the sting. It should be stiff and should transfer loads to the load cells.
3. **Mount the lift load cell vertically.** It should carry vertical loads without side loading.
4. **Mount the drag load cell horizontally.** Align it parallel to the wind direction.
5. **Add hard stops.** Mechanical stops prevent accidental overload if the drone gets bumped or the tunnel speed is too high.
6. **Add the angle-of-attack mechanism.** Use an indexed rotating plate, protractor scale, or inclinometer.
7. **Route wires carefully.** Wires should have slack and strain relief so they do not pull on the moving plate or drone.
8. **Shield the balance.** Keep load cells, electronics, and brackets outside the tunnel flow or covered by a fairing.

16.4 Mounting Rules for Load Cells

- Load the sensor along its intended axis.
- Avoid side loads, twisting, and bending moments on the sensor.
- Do not let the sensor cable carry mechanical load.
- Use rod ends, clevises, or flexures if alignment is not perfect.
- Add overload stops before the load cell reaches its rated limit.
- Calibrate after the entire mechanical structure is assembled.

FUTEK's S-beam manual specifically warns to keep loading flat, centered, and in-line when compensating linkages are not used, and to avoid damaging torque during installation [15]. This is directly relevant to a wind-tunnel balance because off-axis load can create false lift/drag readings.

17 Detailed Calibration Procedure

Calibration should be done after assembly, not before. The load cell alone may behave well on a bench, but the full balance frame introduces alignment, friction, and coupling effects.

17.1 What to Calibrate

You need three levels of calibration:

1. **Single-channel calibration:** raw ADC reading to Newtons for each load cell.
2. **System calibration:** assembled balance response to known lift and drag loads.
3. **Cross-axis calibration:** how much lift leaks into drag and drag leaks into lift.

17.2 Calibration Equipment

Table 9: Calibration equipment.

Item	**Use**
Known masses	Create known forces using $F = mg$
Pulley and string	Convert hanging weight into horizontal drag-axis force
Digital scale	Verify actual mass values
Level / inclinometer	Confirm axes are horizontal/vertical
Clamp stand or fixture	Holds pulley/string at the correct height
Data logger	Records raw and calibrated readings
Notebook / spreadsheet	Stores calibration points and fitted lines

17.3 Lift Calibration, Step by Step

1. Assemble the full balance with the sting and mount attached.
2. Keep the tunnel off.
3. Set the drone or dummy fixture at the test angle.
4. Record the zero reading: $R_{L,0}$.
5. Apply a known vertical load using a mass or hanging fixture.
6. Record the new reading: $R_{L,i}$.
7. Repeat for at least 5 known loads across the expected force range.
8. Fit a line: $F_L = a_L R_L + b_L$.

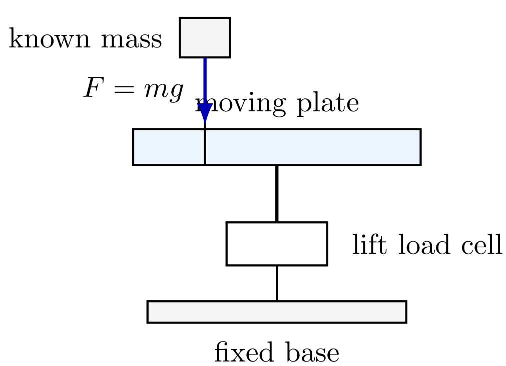


Figure 15: Lift calibration with known vertical loads.

17.4 Drag Calibration, Step by Step

1. Attach a string to the moving plate or sting at the same height as the drag load path.
2. Route the string over a low-friction pulley aligned with the drag axis.
3. Hang a known mass.
4. Record the drag channel reading.
5. Repeat for several masses.
6. Fit a line: $F_D = a_D R_D + b_D$.

The known horizontal force is:

$$F_D = mg \tag{55}$$

If the string is not perfectly horizontal or the pulley has friction, the calibration will be biased. Keep the string aligned with the drag direction.

17.5 Cross-Axis Calibration Procedure

1. Apply a known vertical load only.
2. Record both lift and drag readings.
3. Apply a known horizontal load only.
4. Record both lift and drag readings.
5. Repeat at several force levels.
6. Build a correction matrix if the coupling is significant.

For better data, use a matrix calibration:

$$\begin{bmatrix} L \\ D \end{bmatrix} = \begin{bmatrix} k_{LL} & k_{LD} \\ k_{DL} & k_{DD} \end{bmatrix} \begin{bmatrix} R_L - R_{L,0} \\ R_D - R_{D,0} \end{bmatrix} \tag{56}$$

where k_{LD} and k_{DL} correct for cross-axis coupling.

17.6 Wind-On Test Run

For each wind speed and angle of attack:

1. Set angle of attack.
2. Record tunnel-off tare.
3. Turn on wind tunnel.
4. Wait for speed and force readings to stabilize.
5. Average readings for 5–15 seconds.
6. Subtract tare.
7. Apply calibration slope/matrix.

8. Calculate q, C_L, and C_D.

9. Save raw data and processed data.

Best practice. Keep raw ADC readings, calibrated Newton readings, wind speed, angle of attack, and notes. If something looks wrong later, raw data lets you reprocess instead of retesting.

18 Exact Setup Blueprint: What Goes Where

This section describes the physical setup in a very literal way. In wind-tunnel pictures, the visible post is usually only the **support/sting**. The actual load cells are normally hidden below the tunnel floor, behind the test section, inside a sting housing, or inside a protected balance box. The load cells should not be exposed directly to the airflow.

One-sentence explanation. The wind pushes on the drone; the drone pushes on the sting; the sting pushes on the hidden balance; the hidden balance pushes/pulls on load cells; the electronics turn those load-cell signals into lift and drag.

18.1 Literal Force Path

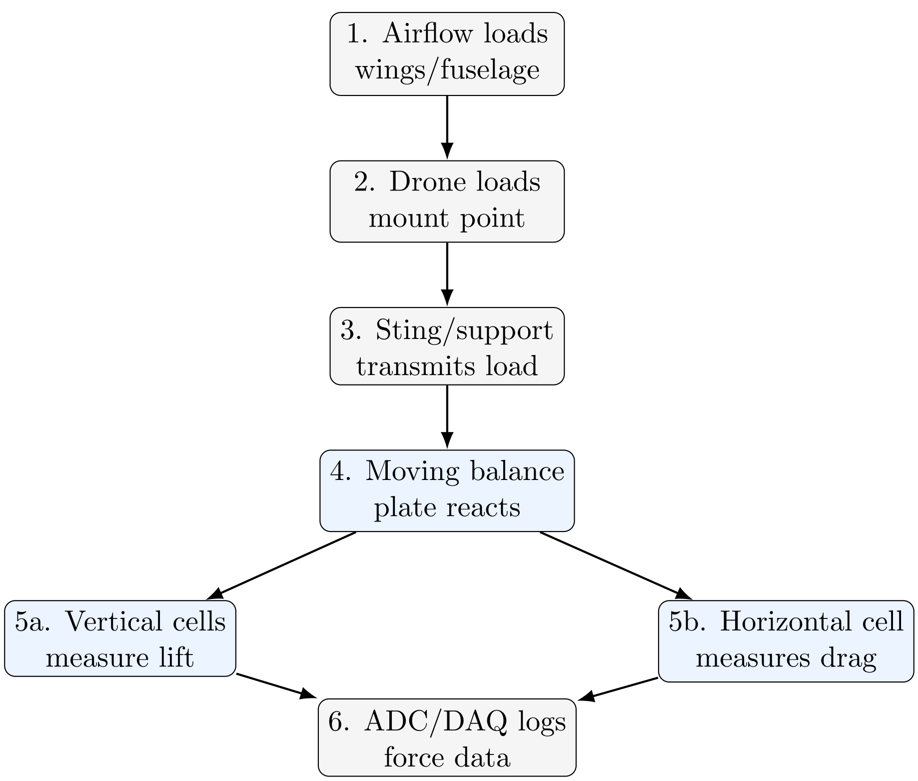


Figure 16: Literal load path from aerodynamic force to recorded data.

18.2 What Bolts to What

A simple student build can be assembled as a stack of mechanical parts:

1. **Drone hard point:** a reinforced belly plate, rear fuselage mount, or internal spar bracket on the drone.

2. **Sting/support rod:** a stiff tube or rod bolted to the drone hard point. This is the only major support piece inside the test section.

3. **Tunnel exit hole/fairing:** the sting passes through a small hole or fairing in the tunnel wall/floor. The hole should not rub the sting.

4. **Angle-of-attack plate:** a rotating plate or hinge outside the flow that sets the drone angle, α.

5. **Moving balance plate:** the plate connected to the sting. This plate transmits loads into the load cells.

6. **Load cells:** mounted between the moving plate and a fixed base frame.

7. **Fixed base frame:** a rigid aluminum/steel frame bolted to the tunnel or table. This is the reference structure.

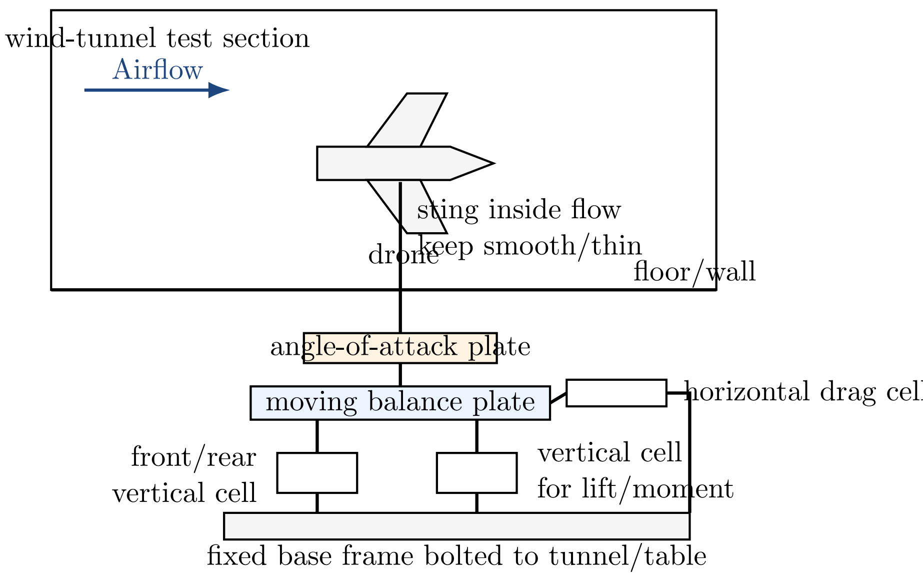


Figure 17: A practical hidden-balance layout. Only the drone and sting are inside the test flow; the load cells and electronics are below or outside the tunnel.

18.3 Good vs. Bad Load-Cell Placement

Table 10: Load-cell placement rules.

Bad placement	**Better placement**
Load cell exposed in the test section	Load cell hidden below floor or inside protected balance box
Sensor wires hanging in the airflow	Wires routed outside the flow with slack and strain relief
Drone mounted directly to a flexible 3D-printed tower	Drone mounted to a stiff sting connected to a metal frame
Drag cell mounted at an angle	Drag cell aligned parallel to wind direction
Lift cell taking side load or twisting	Lift cell loaded vertically through a guided plate or linkage
No overload stops	Mechanical stops before sensor reaches unsafe load

19 Parts List With Specific Buying Strategy

This section gives a practical purchasing plan. Exact stock and prices change, so the safest approach is to buy by **part category and specification**, not only by one product name.

19.1 Core Electronics

Table 11: Electronics needed for the wind-tunnel force balance.

Item	**Example source/search**	**What to check before buying**
Load cell amplifier	SparkFun HX711, SparkFun NAU7802, Phidgets Wheatstone Bridge	Must match bridge load cells. Check sampling rate, noise, supply voltage, and whether it connects by USB, I2C, or GPIO.
Microcontroller/DAQ	Arduino, ESP32, PhidgetBridge, NI DAQ	HX711 works with Arduino/ESP32. Phidgets/DAQ is easier for laptop logging and multi-channel setups.
Lift load cell	Search "S-beam load cell 20 kg" or "50 kg tension compression load cell"	Prefer tension/compression or S-beam for inline loading. Straight-bar scale cells can work but are harder to mount cleanly.
Drag load cell	Search "S-beam load cell 5 kg" or "10 kg tension compression load cell"	Drag is smaller than lift, so do not oversize too much.
Airspeed sensor	Pitot tube + differential pressure sensor such as MS4525DO	Needed for V, q, C_L, and C_D. Tunnel speed measurement is as important as force measurement.

19.2 Mechanical Hardware

Table 12: Mechanical parts for the balance.

Part	Purpose / recommendation
Sting/support tube	Use aluminum or steel tube/rod. Make it stiff and streamlined. Do not let it rub the tunnel wall.
Moving plate	Connects the sting to the load cells. Use aluminum plate or thick reinforced composite/printed structure.
Fixed base	Must be much stiffer than the moving plate. Bolt it to the tunnel/table.
Rod ends/clevis joints	Help apply load axially to S-beam cells without bending them.
Angle plate/protractor	Allows repeatable angle of attack. Use indexed holes or a digital inclinometer.
Pulleys/string	Used for drag calibration with hanging masses.
Calibration masses	Known weights for lift and drag calibration. Verify actual mass if possible.
Hard stops	Protect load cells from accidental overload. Leave small clearance during normal operation.
Fairing/cover	Shields the support opening and balance from airflow.

19.3 Recommended Purchase Setup for This Drone

For a first real build on the current drone size, start with:

- **Lift:** one 50 kg load cell, or two 20 kg to 50 kg vertical cells if also measuring pitch moment.
- **Drag:** one 10 kg load cell.
- **DAQ:** Phidgets bridge/multi-channel bridge if budget allows; otherwise one HX711/NAU7802 board per load cell.
- **Airspeed:** pitot tube and differential-pressure sensor or the wind tunnel's built-in calibrated speed reading.
- **Frame:** aluminum extrusion or steel/aluminum plate; avoid relying only on weak printed plastic for the load path.

Best practical advice. If the project only needs lift and drag, use two cells: one vertical and one horizontal. If the project needs useful fixed-wing stability data, use three cells: two vertical cells for lift plus pitching moment, and one horizontal cell for drag.

20 Load-Cell Selection Worksheet

Use this worksheet before ordering sensors.

20.1 Inputs to Measure

Table 13: Fill-in inputs for load-cell sizing.

Quantity	Symbol	Your value
Total wingspan	b	118.1 in = 3.0 m
Average chord	c_{avg}	0.248 m (9.8 in)
Wing area	$S = bc_{avg}$	$0.744\,\mathrm{m}^2$
Maximum tunnel speed	V_{max}	________ m/s
Estimated max lift coefficient	$C_{L,max}$	1.0–1.4 starting guess
Estimated max drag coefficient	$C_{D,max}$	0.06–0.10 (glider)
Model weight (lift-cell tare)	W	4.7 kg = 46 N
Safety factor	SF	2–3

20.2 Sizing Equations

$$S = bc_{avg} \tag{57}$$

$$q_{max} = \frac{1}{2}\rho V_{max}^2 \tag{58}$$

$$L_{estimate} = q_{max} S C_{L,max} \tag{59}$$

$$D_{estimate} = q_{max} S C_{D,max} \tag{60}$$

$$F_{cell,rated} \geq SF(F_{estimate} + F_{tare}) \tag{61}$$

Then convert Newtons to kilogram-force:

$$F_{kgf} = \frac{F_N}{9.807} \tag{62}$$

20.3 Quick Capacity Table

The table below assumes the real Fox geometry, $S \approx 0.744\,\mathrm{m}^2$, with $C_{L,max} = 1.2$ and $C_{D,max} = 0.08$ (a glider value, profile plus induced drag at high lift).

Table 14: Estimated forces for the FMS 3000 mm Fox ($S = 0.744\,\mathrm{m}^2$).

Tunnel speed	q	Lift estimate	Drag estimate	Suggested cells
10 m/s	61 Pa	55 N / 5.6 kgf	3.6 N / 0.4 kgf	Lift 20 kg, Drag 5 kg
15 m/s	138 Pa	123 N / 12.5 kgf	8.2 N / 0.8 kgf	Lift 20 −−50 kg, Drag 5 kg
20 m/s	245 Pa	219 N / 22.3 kgf	14.6 N / 1.5 kgf	Lift 50 kg, Drag 5 −−10 kg
25 m/s	383 Pa	342 N / 34.9 kgf	22.8 N / 2.3 kgf	Lift 50 kg or dual cells, Drag 10 kg

Interpretation. If your tunnel is around 10 m/s (near this glider's stall), a 20 kg lift cell may be enough. If the tunnel reaches 15 −−20 m/s, use a 50 kg lift channel; remember the lift cells also carry the 46 N model weight as tare. Drag stays small across the whole range (under 25 N), so a 5 −−10 kg drag cell gives both safety and good resolution. The drag forces here are much smaller than for a draggy powered model because the Fox is an efficient, high-aspect-ratio glider.

21 Calibration Worksheet and Data Processing

The goal of calibration is to convert raw ADC readings into Newtons. Calibration must be done with the full support installed because the sting, plate, linkages, and wire routing affect the readings.

21.1 Single-Axis Calibration Table

Use this table once for the lift axis and once for the drag axis.

Table 15: Calibration worksheet for one force channel.

Mass	Known force $F = mg$	Raw reading	Notes
0 g	0 N		zero/tare
100 g	0.981 N		
250 g	2.45 N		
500 g	4.91 N		
1000 g	9.81 N		
2000 g	19.6 N		only if safe

Fit a straight line:

$$F = aR + b \tag{63}$$

where R is the raw ADC reading, a is the scale factor, and b is the offset.

21.2 Cross-Axis Check Table

Table 16: Cross-axis coupling check.

Applied load	Known force	Lift channel reads	Drag channel reads	Problem indicated
Pure lift				Drag reading should be near zero
Pure drag				Lift reading should be near zero
Pure lift, larger				Check if coupling changes with force
Pure drag, larger				Check if coupling changes with force

If cross-axis coupling is small, note it as an uncertainty. If it is large, use the matrix correction described earlier in the notes.

21.3 Wind-Tunnel Data Processing Algorithm

For each angle of attack and tunnel speed:

1. Read raw lift and drag with wind off: $R_{L,off}$, $R_{D,off}$.

2. Turn wind on and wait for steady speed.

3. Average raw readings for 5–15 seconds: $R_{L,on}$, $R_{D,on}$.

4. Subtract wind-off readings:

$$\Delta R_L = R_{L,on} - R_{L,off} \tag{64}$$
$$\Delta R_D = R_{D,on} - R_{D,off} \tag{65}$$

5. Convert to Newtons using calibration:

$$L = a_L \Delta R_L \tag{66}$$
$$D = a_D \Delta R_D \tag{67}$$

6. Calculate dynamic pressure:

$$q = \frac{1}{2}\rho V^2 \tag{68}$$

7. Calculate coefficients:

$$C_L = \frac{L}{qS} \tag{69}$$
$$C_D = \frac{D}{qS} \tag{70}$$

22 Presentation Explanation Script

If explaining the setup verbally, use this simple description:

> The aircraft is mounted on a sting, but the sting is not the sensor. The sting is only the mechanical path that transfers the aerodynamic loads from the aircraft into a hidden force balance. The actual load cells are kept outside the airflow so the wind does not push on the sensors and contaminate the data. One load cell is aligned vertically to measure lift, one is aligned horizontally to measure drag, and a third measurement can be added with two vertical cells to estimate pitching moment. Before testing, the balance is calibrated with known weights so raw electrical readings can be converted into Newtons. During the wind-tunnel run, we subtract the wind-off tare reading from the wind-on reading, then use the measured airspeed to calculate C_L and C_D.

23 Testing Procedure

A basic test sequence should look like this:

1. Mount drone on sting.

2. Connect lift and drag load cells.

3. Connect each load cell to its amplifier/ADC.

4. Connect electronics to Arduino/ESP32 and laptop.

5. Set the drone angle of attack.

6. Record wind-off tare values.

7. Turn tunnel on and set wind speed.
8. Wait for readings to stabilize.
9. Average lift and drag readings over several seconds.
10. Subtract wind-off tare.
11. Calculate C_L and C_D.
12. Repeat for each angle of attack and wind speed.

23.1 Suggested Data Table

Table 17: Recommended data columns for wind-tunnel tests.

α	V	L	D	q	C_L	C_D	Notes
0°							
5°							
10°							
15°							

24 Common Mistakes and How to Avoid Them

Table 18: Common measurement problems in student wind-tunnel balances.

Problem	**What Happens**	**Fix**
Load cell too large	Small forces are hard to resolve	Use expected force estimates before choosing capacity
Load cell too small	Sensor overloads or permanently bends	Use safety factor of about 2–3
Weak mount	Readings vibrate or drift	Use rigid aluminum/steel or strong 3D-printed structure
Bad alignment	Lift appears as drag or drag appears as lift	Align force axes carefully and calibrate both axes
No tare reading	Offset contaminates force results	Tare at each angle before wind-on run
Wires pulling on drone	Fake force readings	Strain-relieve wires and keep them flexible
Support interference	Mount changes airflow and drag	Use thin/streamlined sting and document limitations
Propeller on during basic test	Thrust mixes with drag	Start with propeller off, then run powered tests separately

25 Final Design Conclusion: Selected Force-Balance Architecture

After comparing possible force-sensing methods for the fixed-wing drone wind-tunnel test, the recommended design is a hidden external force-balance system using two bar/beam-type load cells for vertical lift measurement and one S-beam load cell for horizontal drag measurement. This arrangement provides the best balance of accuracy, buildability, cost, and clarity for a student-scale wind-tunnel setup.

The aircraft is the FMS 3000 mm Fox, with a wingspan of 3.0 m (118.1 in), a wing area of 0.744 m^2, and a flying weight of about 4.7 kg. Because this is a relatively large model aircraft, the force-balance system must be strong enough to safely handle significant lift loads while still being sensitive enough to measure smaller drag forces. Lift is expected to be much larger than drag, so the lift and drag axes should not use the same sensor range. The final design therefore uses stronger vertical bar load cells for lift and a smaller, more sensitive S-beam load cell for drag.

The two bar/beam load cells are used for lift because they can support a moving balance plate vertically while measuring the upward or downward force transmitted through the aircraft support sting. Using two vertical load cells instead of one improves the design because the load is distributed over two supports, the balance plate is more stable, and the difference between the two readings can be used to estimate pitching moment. The total lift is found by summing the two vertical load-cell forces:

$$L = F_A + F_B \tag{71}$$

where F_A and F_B are the corrected forces from the front (upstream) and rear (downstream) vertical bar load cells. If the two cells are separated by a known distance d, the pitching moment about the midpoint can be estimated as:

$$M_{pitch} \approx (F_B - F_A)\frac{d}{2} \tag{72}$$

This uses the same sign convention as the three-cell layout earlier in these notes. This matters because a fixed-wing drone does not only produce lift and drag; it also tends to rotate nose-up or nose-down as angle of attack changes.

The S-beam load cell is selected for drag because drag acts primarily in the horizontal airflow direction. An S-beam load cell is better suited for this axis because it is designed to measure inline tension and compression. When connected between the moving balance plate and the fixed frame using threaded rod, clevis joints, or rod-end bearings, the S-beam directly measures the horizontal push or pull created by aerodynamic drag. This creates a cleaner force path than trying to measure drag with a bar load cell mounted in bending.

The load cells should be hidden from the airflow, either below the wind-tunnel floor or behind the test-section wall. This is necessary because exposed sensors, wires, brackets, and electronics would create their own aerodynamic drag and contaminate the measurement. The wind should act only on the aircraft and the smallest possible support sting or fairing. The support sting then transfers the aerodynamic loads into the hidden balance system. In other words, the sensors measure the forces transmitted through the aircraft mount rather than being placed directly in the flow.

The final force path is:

$$\text{Airflow} \rightarrow \text{Drone} \rightarrow \text{Sting/Support Beam} \rightarrow \text{Moving Balance Plate} \rightarrow \text{Load Cells} \rightarrow \text{Data Acquisition} \tag{73}$$

This design is more accurate than mounting sensors directly on the aircraft or placing sensors visibly in the test section because it separates the aerodynamic measurement from sensor drag, wire drag, and mounting interference. The result is a cleaner, more repeatable system capable of measuring lift, drag, and approximate pitching moment.

26 Visual Setup Explanation

26.1 Side View of the Hidden Force Balance

```
Airflow
```

```
----->

          [ Fixed-Wing Drone ]
                   |
                   |
            Angle-of-Attack
                Bracket
                   |
                   |
             Sting / Support
                   |
    Wind Tunnel Floor or Wall
                   |
                   v
          Moving Balance Plate
      +----------------------------+
      |                            |
      +-----+---------------+-----+
            |               |
            |               |
    Bar Load Cell A   Bar Load Cell B
      Vertical Lift     Vertical Lift
            |               |
            v               v
       Fixed Base Frame / Tunnel Structure
```

In this side-view layout, the aircraft is mounted inside the test section, but the load cells are hidden below the tunnel floor. The visible sting only transfers the aircraft loads into the hidden force-balance structure.

26.2 Top View of Drag Measurement Using an S-Beam

```
Airflow
----->

              [ Drone and Sting ]
                       |
                       v
          +--------------------------+
          |  Moving Balance Plate    |
          +--------------------------+
                       |
                       | horizontal drag load path
                       v
   Moving Plate -- Rod/Clevis -- S-Beam -- Rod/Clevis -- Fixed Frame
```

The S-beam is placed horizontally and inline with the expected drag direction. When the wind pushes the aircraft backward, the moving balance plate pushes or pulls on the S-beam. The S-beam reading is then converted into drag force.

26.3 Force Path Diagram

```
Airflow
   |
   v
Pushes on drone
   |
   v
Drone loads the sting
   |
   v
Sting loads the moving balance plate
   |
   +----------------------+
   |                      |
   v                      v
Vertical bar cells   Horizontal S-beam
measure lift         measures drag
   |                      |
   +----------+-----------+
              |
              v
Data system converts sensor signals into Newtons
```

26.4 Good Setup vs. Bad Setup

```
GOOD SETUP:

Airflow -----> [ Drone ]
                   |
              Smooth sting
                   |
        Load cells hidden below floor

Only the aircraft and small sting are exposed to airflow.
The sensors measure transmitted aircraft loads.

BAD SETUP:

Airflow -----> [ Drone ] -- exposed load cell -- wires -- bracket

The airflow pushes on the sensor, wires, and bracket.
The measurement includes fake drag from the test hardware.
```

27 Installation and Build Instructions

27.1 Step 1: Build the Fixed Base Frame

The fixed base frame is the non-moving structure that supports the entire force-balance assembly. It should be rigid and heavy enough that it does not flex during testing. Aluminum extrusion, steel plate, or a thick machined/bolted frame is preferred. Wood can be used for early prototyping, but it may flex and reduce measurement quality.

The fixed frame should be attached to the wind tunnel structure or to a heavy independent base. The frame must provide mounting points for the two vertical bar load cells and the horizontal S-beam load cell.

27.2 Step 2: Install the Two Bar Load Cells

Each bar/beam load cell has a fixed side and a load side. The fixed side should bolt to the base frame. The load side should connect to the moving balance plate. The bar cells must not be fully clamped along their entire length because they need to bend slightly to measure force.

```
Correct bar load-cell mounting:

       Moving Balance Plate
                |
                v
        Load end of bar cell
      +-----------------------+
      |       Load Cell       |
      +-----------------------+
        Fixed end of bar cell
                |
                v
          Fixed Base Frame
```

The two bar load cells should be spaced apart along the aircraft's pitch direction. This allows the system to measure total lift and estimate pitching moment.

27.3 Step 3: Install the Moving Balance Plate

The moving balance plate is the intermediate plate between the aircraft support sting and the load cells. The drone does not bolt directly to the load cells. Instead, the drone bolts to the sting, the sting bolts to the moving plate, and the moving plate loads the sensors.

The moving plate should be stiff enough that it does not bend significantly. Aluminum plate is recommended. If the plate bends, the readings from the load cells may become inconsistent or nonlinear.

27.4 Step 4: Attach the Sting or Support Post

The sting/support post connects the aircraft to the moving balance plate. For the first version of the design, a vertical belly-mounted sting is the simplest to build. A rear sting is more aerodynamically clean but harder to attach to a full fixed-wing drone.

The sting should be:

- strong enough to hold the aircraft without vibration,
- as thin as practical to reduce aerodynamic interference,
- aligned with the balance plate,
- removable for maintenance and calibration.

27.5 Step 5: Add an Angle-of-Attack Adjustment Bracket

An angle-of-attack bracket should be placed between the drone and the sting, or between the sting and moving plate. This bracket allows the drone to be tested at multiple angles such as:

$$\alpha = -5^\circ,\ 0^\circ,\ 5^\circ,\ 10^\circ,\ 15^\circ,\ 20^\circ \tag{74}$$

The bracket should lock firmly at each angle. Any looseness will create vibration and measurement noise.

27.6 Step 6: Install the S-Beam Load Cell Horizontally

The S-beam load cell should be mounted horizontally between the moving balance plate and the fixed frame. It should be inline with the airflow direction. Use threaded rod, clevis joints, or rod-end bearings so the S-beam is loaded in pure tension/compression.

```
Top view:

Moving Plate -- Clevis/Rod -- S-Beam -- Clevis/Rod -- Fixed Frame
```

The S-beam should not be twisted, side-loaded, or bent. Misalignment will cause inaccurate drag readings and may damage the sensor.

27.7 Step 7: Wire the Load Cells

Each load cell needs its own amplifier or data-acquisition channel. With two bar load cells and one S-beam load cell, the system needs three load-cell channels.

Possible electronics options include:

- HX711 amplifier boards,
- NAU7802 load-cell amplifier boards,
- Phidgets bridge interface,
- dedicated laboratory DAQ with bridge input.

For a simple Arduino or ESP32 setup, one amplifier board per load cell is acceptable. For a cleaner laboratory setup, a multi-channel bridge interface is preferred.

27.8 Step 8: Add Wire Strain Relief

The sensor wires must be secured so they do not pull on the moving balance plate. If the wires are tight, they can create fake forces in the lift or drag readings.

Use:

- zip-tie anchors,
- flexible cable loops,
- cable glands,
- strain-relief clips.

Wires and electronics should remain outside the airflow whenever possible.

27.9 Step 9: Add Mechanical Overload Stops

Mechanical stops should be installed to prevent the moving plate from overloading the sensors. These stops should allow small motion but prevent excessive deflection if the aircraft is bumped or the wind speed is too high.

```
      Moving Plate
           |
           | small allowed gap
           v
   Overload Stop / Hard Limit
```

This protects the load cells from permanent damage.

27.10 Step 10: Check Alignment Before Testing

Before any wind-tunnel run:

- verify that the sting is vertical or aligned as intended,
- verify that the S-beam is horizontal and inline with airflow,
- verify that nothing rubs on the moving plate,
- verify that the load cells are not preloaded beyond safe limits,
- verify that wires are not pulling on the system,
- verify that the aircraft is centered in the test section.

28 Calibration Procedure (Final Design)

28.1 Lift Calibration

Lift calibration is done by applying known vertical loads to the moving balance plate or sting. Known masses are converted into force using:

$$F = mg \tag{75}$$

where m is mass in kilograms and $g = 9.807\,\mathrm{m/s^2}$.
Example calibration forces:

$$1\,\mathrm{kg} = 9.81\,\mathrm{N} \tag{76}$$
$$2\,\mathrm{kg} = 19.6\,\mathrm{N} \tag{77}$$
$$5\,\mathrm{kg} = 49.0\,\mathrm{N} \tag{78}$$
$$10\,\mathrm{kg} = 98.1\,\mathrm{N} \tag{79}$$

Table 19: Lift calibration worksheet for the two vertical bar cells.

Mass Applied	**Force Applied**	**Bar Cell A Raw**	**Bar Cell B Raw**
0 kg	0 N		
1 kg	9.81 N		
2 kg	19.6 N		
5 kg	49.0 N		
10 kg	98.1 N		

A calibration curve is then generated for each cell:

$$F = aR + b \tag{80}$$

where R is the raw sensor reading, a is the calibration slope, and b is the offset.

28.2 Drag Calibration

Drag calibration is done using a string, pulley, and hanging mass to apply a known horizontal force to the moving plate.

```
Moving Plate -- String -- Pulley
                            |
                            |
                          Mass
```

The hanging mass creates a known horizontal force:

$$F = mg \tag{81}$$

Example drag calibration values:

Table 20: Drag calibration worksheet for the S-beam cell.

Mass Applied	**Horizontal Force**	**S-Beam Raw Reading**
0 g	0 N	
100 g	0.981 N	
200 g	1.96 N	
500 g	4.91 N	
1000 g	9.81 N	

28.3 Wind-Off Tare and Wind-On Correction

The wind-off reading must be recorded before each test condition. This removes the effects of aircraft weight, sensor preload, mounting bias, and small alignment errors.

Corrected lift is:

$$L = (F_{A,on} + F_{B,on}) - (F_{A,off} + F_{B,off}) \tag{82}$$

Corrected drag is:

$$D = F_{D,on} - F_{D,off} \tag{83}$$

Corrected pitching moment is:

$$M_{pitch} \approx \left[(F_B - F_A)_{on} - (F_B - F_A)_{off} \right] \frac{d}{2} \tag{84}$$

28.4 Cross-Axis Coupling Check

Cross-axis coupling occurs when a pure lift load creates a false drag reading, or when a pure drag load creates a false lift reading. This must be checked during calibration.

A simple correction test is:

1. Apply a known vertical force and record all three channels.
2. Apply a known horizontal force and record all three channels.
3. Record how much each non-target channel changes.
4. Use this information to subtract coupling errors from the final data.

For more advanced correction, a calibration matrix can be used:

$$\begin{bmatrix} L \\ D \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} R_L \\ R_D \end{bmatrix} \tag{85}$$

29 Equations and Data Processing (Final Design)

The main aerodynamic equations are:

$$q = \frac{1}{2}\rho V^2 \tag{86}$$

$$L = qSC_L \tag{87}$$

$$D = qSC_D \tag{88}$$

Solving for coefficients:

$$C_L = \frac{L}{qS} \tag{89}$$

$$C_D = \frac{D}{qS} \tag{90}$$

where:

- q is dynamic pressure,
- ρ is air density,
- V is wind-tunnel speed,
- S is wing reference area,
- C_L is lift coefficient,
- C_D is drag coefficient.

The reference wing area is estimated by:

$$S = b\bar{c} \tag{91}$$

where b is wingspan and $\bar{c}$ is average chord.
For this aircraft, use the full wingspan:

$$b \approx 118.1\,\text{in} \approx 3.0\,\text{m}, \quad S \approx 0.744\,\text{m}^2 \tag{92}$$

The force-balance readings are converted into aerodynamic forces using:

$$L = F_A + F_B \tag{93}$$

$$D = F_{Sbeam} \tag{94}$$

after wind-off tare correction.

30 Choosing the Correct Load Cells (Final Design)

Load-cell capacity should be selected by estimating the largest expected force, then applying a safety factor. The main steps are:

1. Estimate full wingspan b.
2. Estimate average wing chord $\bar{c}$.
3. Calculate wing area $S = b\bar{c}$.
4. Choose the maximum wind-tunnel speed V.
5. Estimate reasonable values for C_L and C_D.
6. Calculate expected lift and drag.
7. Apply a safety factor.
8. Select load cells with enough capacity but not excessive capacity.

The estimated lift is:

$$L_{max} = \frac{1}{2}\rho V^2 S C_{L,max} \tag{95}$$

The estimated drag is:

$$D_{max} = \frac{1}{2}\rho V^2 S C_{D,max} \tag{96}$$

The required load-cell capacity is:

$$F_{rated} \geq SF \cdot F_{max} \tag{97}$$

where SF is the safety factor. Consistent with the rest of these notes, a student wind-tunnel balance should use:

$$SF = 2 \text{ to } 3 \tag{98}$$

Choosing a load cell that is too small risks overload and permanent damage. Choosing a load cell that is too large reduces resolution because the expected force uses only a small portion of the sensor range.

For this drone, the recommended starting ranges are:

Table 21: Final-design recommended load-cell ranges.

Measurement	**Sensor Type**	**Recommended Range**	**Notes**
Lift	Two bar/beam load cells	20 kg each or 50 kg each	Depends on speed and chord
Drag	One S-beam load cell	5 kg or 10 kg	Smaller range gives better drag resolution
Pitching moment	Difference between lift cells	Same as lift cells	Requires known spacing d

For a safer first build, use two 20 kg bar load cells for lift and one 10 kg S-beam for drag. If calculations show high lift at maximum tunnel speed, increase the lift sensors to 50 kg each.

31 Parts List and Buying Guide (Final Design)

Table 22: Final-design bill of materials.

Part	Purpose	Recommended Specs	Qty	Example Sources	Notes
Bar/beam load cell	Measures vertical lift force	20 kg or 50 kg capacity	2	SparkFun, Phidgets, Omega, DigiKey	Used under moving balance plate
S-beam load cell	Measures horizontal drag force	5 kg or 10 kg capacity	1	FUTEK, Omega, Phidgets, DigiKey	Mount inline with airflow direction
Load-cell amplifier/ADC	Converts small bridge signal to digital data	HX711, NAU7802, Phidgets Bridge, or DAQ	3 ch.	SparkFun, Phidgets	One channel per load cell
Microcontroller or DAQ	Logs data	Arduino, ESP32, Phidgets USB, or lab DAQ	1	SparkFun, Phidgets, NI	DAQ is cleaner for multi-channel testing
Pitot tube / airspeed sensor	Measures wind speed	Differential pressure sensor or UAV airspeed kit	1	Matek, Holybro, lab suppliers	Needed for C_L and C_D
Fixed base frame	Holds sensors and balance assembly	Aluminum extrusion, steel frame, or rigid plate	1	McMaster, 80/20	Must not flex significantly
Moving balance plate	Transfers loads from sting to sensors	Rigid aluminum plate	1	McMaster, machine shop	Should be stiff and flat
Sting/support post	Holds drone in tunnel	Metal tube/rod with fairing if possible	1	Hardware supplier, machine shop	Keep small and rigid
Angle-of-attack bracket	Allows aircraft pitch adjustment	Locking bracket with angle marks	1	Custom / 3D printed / machined	Must lock tightly
Clevis joints / rod ends	Align S-beam load path	Match S-beam thread size	2–4	McMaster, FUTEK, Omega	Prevents bending of S-beam
Threaded rod	Connects S-beam to plate/frame	Match sensor threads	1–2	McMaster, hardware store	Used for drag linkage
Bolts, spacers, washers	Mechanical assembly	Manufacturer-recommended sizes	As needed	Hardware supplier	Use correct torque
Mechanical stops	Protect sensors from overload	Adjustable hard stops	2–4	Custom / hardware supplier	Prevents accidental overloading
Wire strain relief	Prevents wires from creating fake loads	Cable clips, glands, zip mounts	As needed	Hardware supplier	Keep wires from pulling on moving plate

32 Reference Images and Guides to Include

The final report should include reference images or recreated schematic diagrams based on the following types of sources:

- NASA wind-tunnel balance diagrams showing how aircraft models are mounted to balances.

- NASA Glenn force-balance examples showing lift, drag, and moment measurement concepts.
- Educational wind-tunnel balance examples showing load-cell-based lift and drag measurement.
- Manufacturer diagrams showing S-beam load cells mounted with clevises and rod-end bearings.
- Load-cell electronics guides showing Wheatstone bridge wiring and amplifier/ADC connections.

Any image used directly should be credited. If copyright or usage is unclear, the final report should use original recreated diagrams based on the concepts rather than directly copying images.

33 Final Presentation Explanation

The final selected design uses a hidden external force balance. The drone is mounted on a support sting inside the wind tunnel, but the load cells are placed outside the airflow. The sting transfers all aerodynamic loads from the drone into a moving balance plate. Two vertical bar load cells underneath the plate measure lift, and because they are spaced apart, their difference can also estimate pitching moment. A horizontal S-beam load cell connects the moving plate to the fixed frame and measures drag through inline tension or compression.

This design is preferred because it keeps the sensors and wiring out of the airflow, reducing false drag and measurement interference. It also separates the lift and drag axes mechanically, which makes the data easier to calibrate and interpret. After calibration with known weights, the raw sensor readings can be converted into lift and drag forces in Newtons. Using the measured wind speed and wing reference area, the final data can then be converted into nondimensional aerodynamic coefficients C_L and C_D.

34 Summary

For this fixed-wing drone wind-tunnel project, the best starting design is a **two-axis external load-cell force balance**, extendable to a three-cell balance for pitching moment. Use a vertical load cell (or two spaced vertical cells) for lift, a horizontal S-beam load cell for drag, a rigid sting mount, and an adjustable angle-of-attack mechanism. Convert raw readings into Newtons through calibration, subtract wind-off tare, and calculate aerodynamic coefficients using $C_L = L/(qS)$ and $C_D = D/(qS)$.

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